

Intelligent Wristband Based on ESP32

Project Overview

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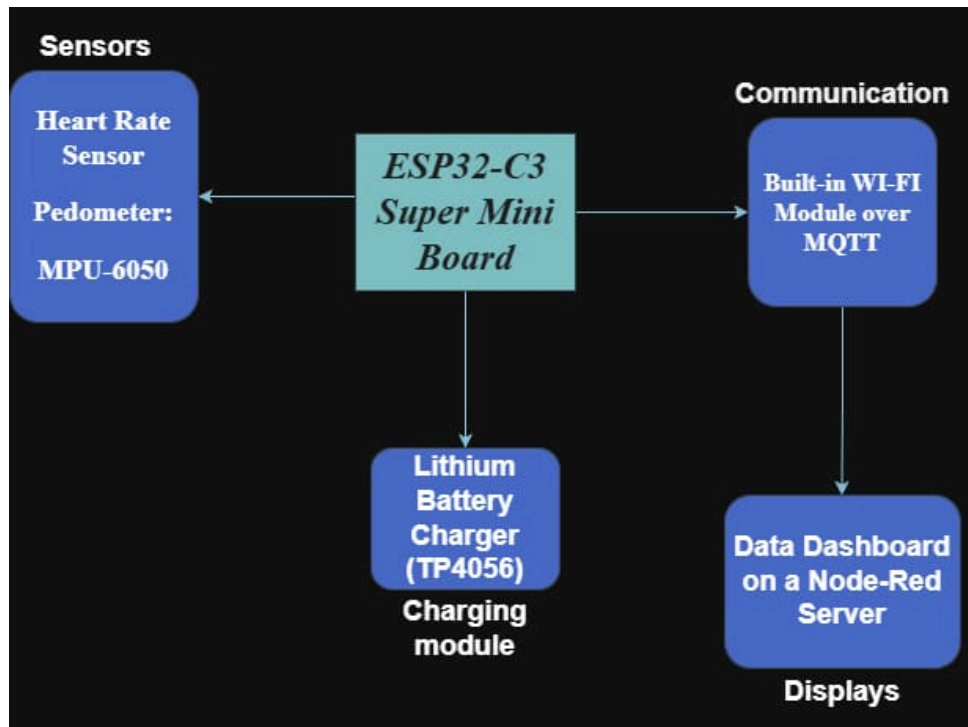


Figure 1: Components of the prototype.

The project monitors mainly pulse rate and burned calories using two sensors: A MAX30102 Heart Rate sensor through photoplethysmography (PPG), which detects changes in blood volume by measuring light absorption in skin, interfaced through I2C, and an MPU-6050/ADXL345 gyroscope/accelerometer that functions as a pedometer (step counter) through evaluating the received angular accelerations as movement (steps) and calculating the equivalent caloric consumption.

Sensors are controlled via the ESP32-C3 Super Mini for its small form-factor, power consumption, and built-in WiFi module powered using a 3.7V rechargeable Li-ion battery regulated using a TP4056 regulator.

Considering future smart-home applications, the processed data is sent using the MQTT protocol due to its publish/subscribe model. A Node-Red dashboard server is used to listen on the MQTT channel and trigger warnings/log diagnostics based on the received vital data.

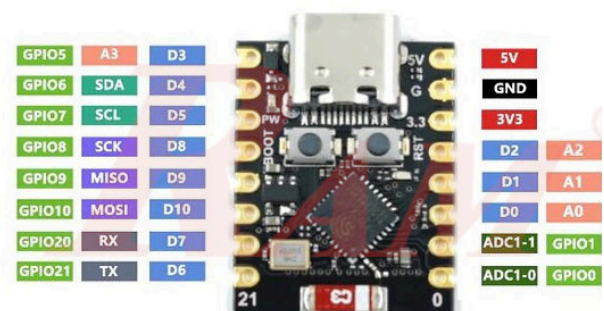


Figure 2: The ESP32-C3 Super Mini Board.

The prototype will consist of a small housing on a wristband holding the battery and a small PCB with the MCU and sensors. The possibility of removing the ESPRESSIF-based chip and routing it directly on the PCB will be investigated after having a working V1 on a perfboard.